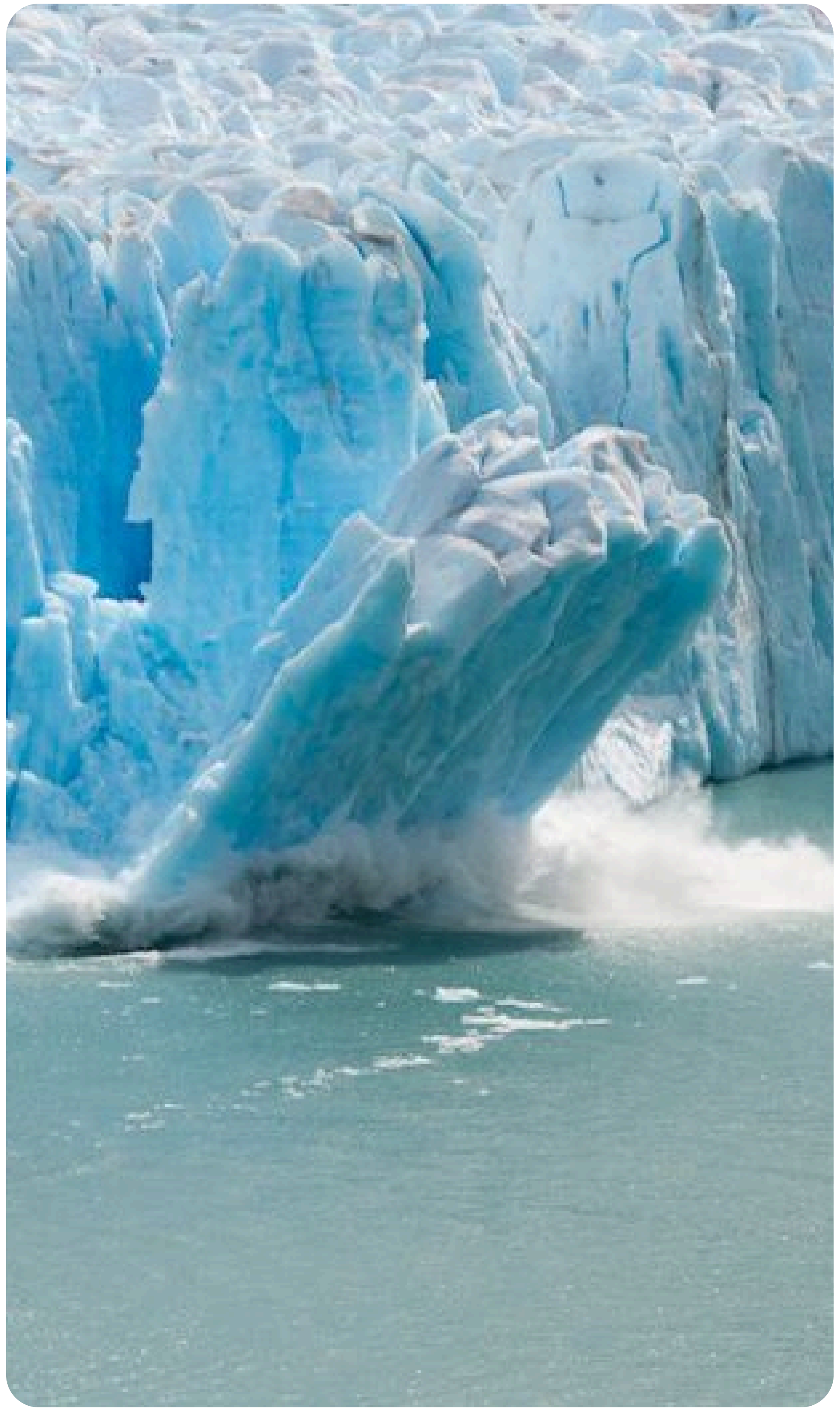
HOW DOES GLACIER CALVING ACCELERATE THE MELTING OF ICE?

Study of a new method designed to understand the phenomenon



GLACIER CALVING

- When large pieces of ice break off the front of a glacier and fall into the ocean
- Natural process but major factor in the fast reduction of the ice sheet
- Interaction with warm seawater => accelerates glacier melting

THE STUDY

- University of Zurich X University of Washington
- Fiber-optic technology
- Goal: to monitor what happens underwater when the ice melts

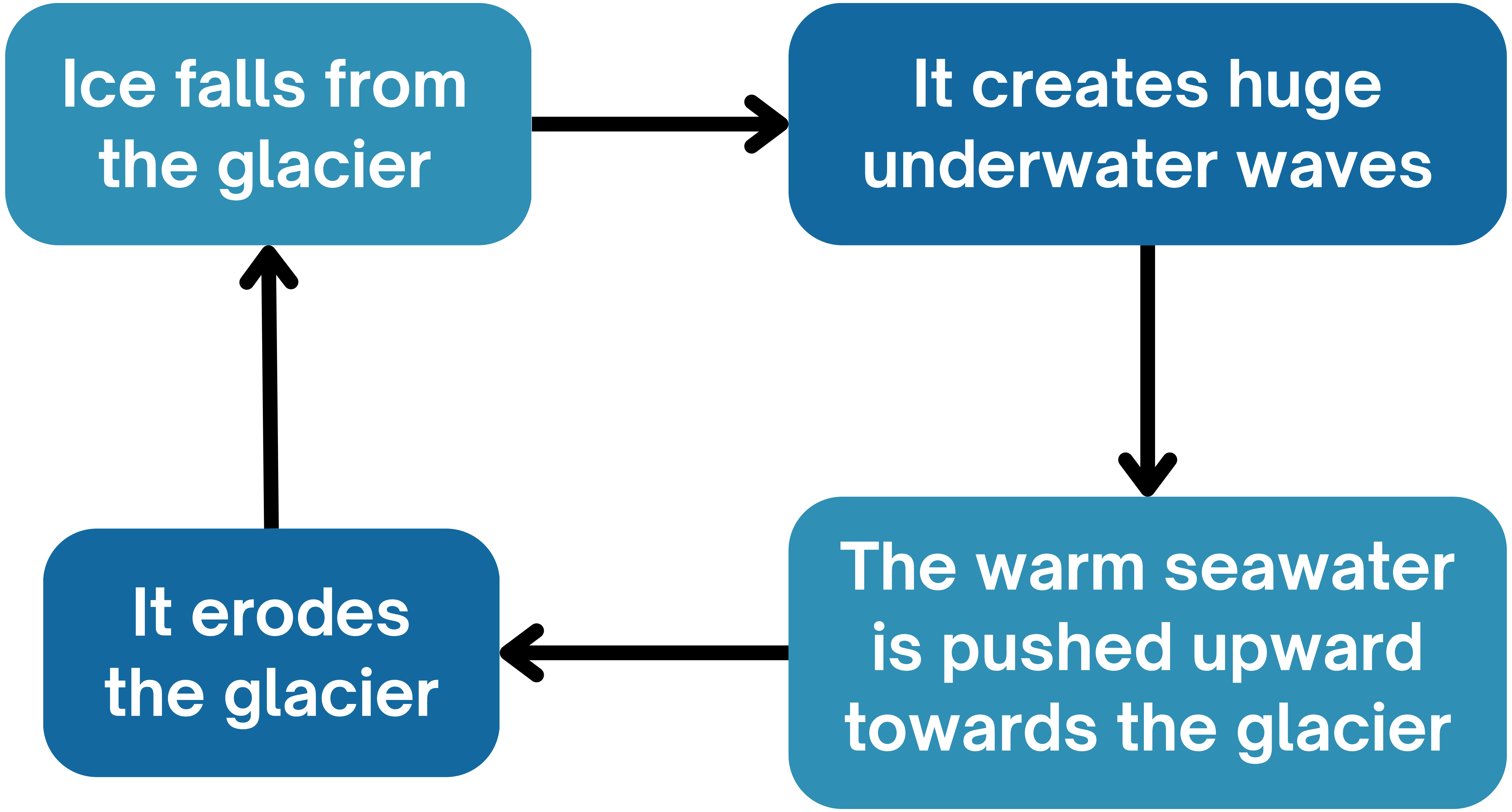
Eqalorutsit Kangilliit Sermiat:
- fast-moving glacier in Southern Greenland
- releases 3.6 km^3 of ice into the ocean every year

10km long fiber-optic cable on the seafloor
Used technique: Distributed Acoustic Sensing (DAS)
=> Detection of tiny vibrations along the cable

Vibrations sources: crevasses forming, ice blocks falling, waves, temperature changes...

WHAT THEY FOUND

The calving multiplier effect



IMPLICATIONS

- If Greenland melted completely, global sea level would rise by around 7 meters
- Meltwater entering the ocean can disrupt major currents such as the Gulf Stream
- The fragile ecosystems of Greenland’s fjords can be affected

CONCLUSION

- Calving: **not just a consequence** of climate warming, it also **reinforces** it
- The new fiber-optic technology: gives scientists essential data
- Reminder of how fragile the Earth's cryosphere is

THANK YOU